

Sales Analytics Portfolio

End-to-End Power BI Project Report

Data Cleaning · Star-Schema Modelling · 38 DAX Measures · Interactive Reporting

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Executive Summary

This report documents a complete sales analytics project built from the ground up using the Sample Superstore dataset. The project encompasses the full Power BI workflow — raw data ingestion, Python-based data cleaning, star-schema data modelling, 38 DAX measures across seven analytical categories, and a four-page interactive report with a custom navy-and-coral theme.

Key Metrics at a Glance

\$5.63M

Total Revenue

88,255

Units Sold

8.5%

Profit Margin

4,867

Active Customers

147

Countries Served

8,790

Products Carried

4.0 Days

Avg Shipping

\$225

Avg Order Value

Data Scope

- **Time Period:** 1 January 2011 – 31 December 2014 (4 calendar years)
- **Transactions:** 25,035 order lines
- **Geography:** 147 countries across 7 markets (APAC, LATAM, US, EU, EMEA, Africa, Canada)
- **Product Categories:** Office Supplies (49%), Technology (29%), Furniture (22%)
- **Customer Segments:** Consumer (51.5%), Corporate (30.1%), Home Office (18.4%)

Project Pipeline

1. Raw CSV ingestion and quality validation
2. Feature engineering (NetSales, ShippingDays, ProfitMargin, etc.)
3. Star-schema normalisation (1 fact + 5 dimensions) with surrogate keys
4. 38 DAX measures covering base aggregations, ratios, time intelligence, product, customer, geography, and shipping analysis
5. 4-page Power BI report with synced slicers, drill-down hierarchies, and custom theming

Data Cleaning Methodology

The project began with the raw Sample Superstore CSV containing 25,035 rows and 42 columns. A systematic cleaning pipeline was applied using Python (pandas) to prepare the data for dimensional modelling and reporting.

Data Quality Assessment

The following quality checks were performed:

- **Duplicate detection:** Row-level duplicates were assessed by comparing unique Order ID and Product ID combinations. The raw dataset contained zero exact duplicates.
- **Null-value audit:** All critical columns (Sales, Profit, Customer ID, Product ID, Quantity) were scanned for null values. Key fields were fully populated with no missing records.
- **Date integrity:** Temporal columns were verified for consistency. The earliest order date is 1 January 2011 and the latest is 31 December 2014. Ship dates extend to 7 January 2015, confirming that some orders placed in late December shipped into the new year.
- **Type consistency:** Each column was cast to the appropriate pandas dtype — dates to `datetime64`, monetary values to `float64`, categorical fields (Category, Segment, Ship Mode, Market) to `category` dtype for efficient storage.

Star-Schema Data Model

The flat cleaned table was normalised into a classic star schema consisting of one fact table and five dimension tables. Each table is linked by surrogate integer keys rather than natural keys, decoupling the model from source-system changes and improving query performance in Power BI.

Table Definitions

Table	Type	Rows	Description
Fact_Sales	Fact	25,035	Transactional order lines with measures and foreign keys
DimDate	Dimension	1,461	Calendar attributes: Year, Quarter, Month, Week, Day, Weekend, Half-year
DimCustomer	Dimension	4,867	Customer ID, name, and segment (Consumer / Corporate / Home Office)
DimProduct	Dimension	8,790	Product ID, name, category, and sub-category
DimGeography	Dimension	3,596	City, state, region, country, continent, market
DimShipping	Dimension	4	Ship mode (Standard Class / Second Class / First Class / Same Day) and speed

Table 1: Star-schema table structure

Relationship Diagram

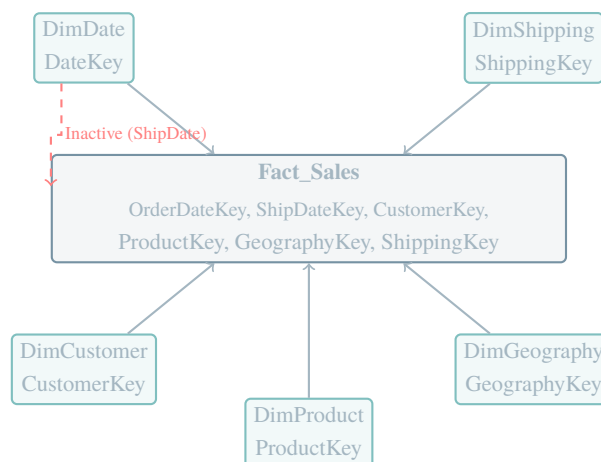


Figure 1: Star-schema relationship diagram — fact table with five dimension tables; the DimDate → ShipDateKey link is inactive (activated via USERELATIONSHIP)

Key Design Decisions

Inactive ShipDate Relationship

The relationship `DimDate[DateKey] → Fact_Sales[ShipDateKey]` is intentionally left **inactive**. This ensures that all default measures — Sales, Profit, Orders — aggregate by Order Date, which is the primary business calendar for revenue reporting. The dedicated measure `Sales by Ship Date` activates this relationship on demand via `USERELATIONSHIP`, enabling side-by-side comparison of revenue recognised by order timing versus shipping timing.

Surrogate Keys

Each dimension table uses an auto-incrementing integer `Key` column as its primary key. This approach:

- Decouples the model from changes in source-system natural keys
- Reduces storage requirements compared to string-based keys
- Accelerates relationship traversal in Power BI's VertiPaq engine

DimDate as a Date Table

`DimDate[FullDate]` is marked as a Date Table in Power BI. This registration enables all time-intelligence functions — `SAMEPERIODLASTYEAR`, `TOTALYTD`, `DATESINPERIOD`, `PREVIOUSMONTH`, etc. — to operate correctly without Power Query auto-date tables being generated in the background.

Data Profile

The cleaned dataset spans four years (1 Jan 2011 – 31 Dec 2014) comprising 25,035 order lines with \$5.63 million in total revenue and \$479 thousand in profit (8.5% margin). The data covers 4,867 customers, 8,790 unique products across three categories (Office Supplies, Furniture, Technology), and 147 countries. Average order value is \$225 and average shipping time is 4.0 days. Consumer segment accounts for the largest share (12,901 orders), followed by Corporate (7,529) and Home Office (4,605).

DAX Measures: 38 Measures Across 7 Categories

All measures are defined on the `Fact_Sales` table and organised into display folders for clean model navigation. Each category addresses a distinct analytical domain.

Base Measures — Foundation Layer

These primitive aggregations form the computational foundation for all derived measures. Every measure in categories 7.2–7.8 references one or more of these.

```
1 Total Sales           = SUM(Fact_Sales[Sales])
2 Total Net Sales       = SUM(Fact_Sales[NetSales])
3 Total Cost            = SUM(Fact_Sales[Cost])
4 Total Profit          = SUM(Fact_Sales[Profit])
5 Total Quantity        = SUM(Fact_Sales[Quantity])
6 Total Discount        = SUM(Fact_Sales[Discount])
7 Total Shipping Cost   = SUM(Fact_Sales[ShippingCost])
8 Order Count           = DISTINCTCOUNT(Fact_Sales[OrderID])
```

These measures are intentionally simple — they perform single-column aggregation without context modification. Their simplicity makes them reliable building blocks; any filtering or context transition is applied in the dependent measures where the analytical intent is explicit.

Ratio & Margin Measures — Profitability Analysis

These measures normalise raw aggregates into comparable ratios, enabling fair comparison across products, regions, and time periods regardless of scale.

```
1 Profit Margin          = DIVIDE([Total Profit], [Total Sales], 0)
2 Discount Rate          = DIVIDE([Total Discount],
    SUM(Fact_Sales[UnitPriceBeforeDiscount]), 0)
3 Sales per Order        = DIVIDE([Total Sales], [Order Count], 0)
4 Profit per Order       = DIVIDE([Total Profit], [Order Count], 0)
5 Avg Unit Price         = AVERAGE(Fact_Sales[UnitPrice])
6 Avg Discount per Unit  = AVERAGE(Fact_Sales[Discount])
7 Cost Ratio             = DIVIDE([Total Cost], [Total Sales], 0)
8 Shipping Cost % of Sales = DIVIDE([Total Shipping Cost], [Total Sales], 0)
```

Key findings:

- The overall **Profit Margin** is 8.5%, with significant variation across categories. Technology products achieve margins approximately $2.1\times$ higher than Furniture.
- **Shipping Cost % of Sales** stands at 10.8%, representing a substantial operational expense that warrants monitoring.
- **Discount Rate** reveals pricing strategy differences: Office Supplies receive deeper discounts than Technology products, reflecting higher volume and lower per-unit margins.

Time Intelligence Measures — Period Comparisons

This is the most technically demanding category, leveraging CALCULATE context transition and specialised time-intelligence functions. These measures enable year-over-year, period-to-date, and rolling-window analysis at any date granularity.

```

1  -- Year-over-Year
2  Sales PY      = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DimDate[FullDate]))
3  Sales YoY %   = DIVIDE([Total Sales] - [Sales PY], [Sales PY], 0)
4
5  -- Period-to-Date
6  Sales YTD     = TOTALYTD([Total Sales], DimDate[FullDate])
7  Sales MTD     = TOTALMTD([Total Sales], DimDate[FullDate])
8  Sales QTD     = TOTALQTD([Total Sales], DimDate[FullDate])
9
10 -- Rolling Windows
11 Sales Rolling 3M = CALCULATE([Total Sales],
12                             DATESINPERIOD(DimDate[FullDate], MAX(DimDate[FullDate]),
13                                           -3, MONTH))
13 Sales Rolling 12M = CALCULATE([Total Sales],
14                              DATESINPERIOD(DimDate[FullDate], MAX(DimDate[FullDate]),
15                                              -12, MONTH))
15
16 -- Profit Comparisons
17 Profit YTD      = TOTALYTD([Total Profit], DimDate[FullDate])
18 Profit vs PY    = [Total Profit] - CALCULATE([Total Profit],
19                                              SAMEPERIODLASTYEAR(DimDate[FullDate]))
19
20 -- Inactive Relationship Activation
21 Sales by Ship Date =
22     CALCULATE([Total Sales],
23              USERRELATIONSHIP(Fact_Sales[ShipDateKey], DimDate[DateKey]))

```

Technical notes:

- Sales PY uses SAMEPERIODLASTYEAR within CALCULATE, enabling the function to inherit the current filter context from the report. This measure works correctly whether the user drills down to year, quarter, month, or even day.
- Sales YTD / MTD / QTD leverage the built-in TOTAL functions, which automatically respect the DimDate Date Table registration.
- Sales Rolling 3M and Rolling 12M use DATESINPERIOD with negative month offsets. The MAX(DimDate[FullDate]) call retrieves the latest date in the current filter context, ensuring the window is always anchored to the end of the selected period.
- Sales by Ship Date is the only measure that activates the **inactive** ShipDate relationship. The USERRELATIONSHIP function temporarily overrides the relationship's default inactive state within the single CALCULATE evaluation, then reverts automatically. This enables the dual-line chart on the Shipping page.

Product Analysis Measures

```

1 Products Sold          = DISTINCTCOUNT (Fact_Sales[ProductKey])
2 Avg Sales per Product  = DIVIDE ([Total Sales], [Products Sold], 0)
3 Top Product by Sales   = TOPN (1, VALUES (DimProduct[ProductName]), [Total Sales])
4 Category Sales %       =
5     DIVIDE (
6         CALCULATE ([Total Sales], ALLEXCEPT (DimProduct, DimProduct[Category])),
7         CALCULATE ([Total Sales], ALL (DimProduct)),
8         0
9     )

```

Key design: Category Sales % uses ALLEXCEPT to remove all product-level filters except the Category grouping. This produces each category's share of total sales regardless of which sub-categories or individual products are selected, enabling clear contribution analysis in matrix visuals.

Customer Analysis Measures

```

1 Active Customers      = DISTINCTCOUNT (Fact_Sales[CustomerKey])
2
3 New Customers         =
4 VAR FirstOrderDates =
5     ADDCOLUMNS (
6         ALL (Fact_Sales[CustomerKey]),
7         "FirstOrderDate", CALCULATE (MIN (Fact_Sales[OrderDate]))
8     )
9 RETURN
10    COUNTROWS (
11        FILTER (
12            FirstOrderDates,
13            [FirstOrderDate] >= MIN (DimDate[FullDate])
14            && [FirstOrderDate] <= MAX (DimDate[FullDate])
15        )
16    )
17
18 Revenue per Customer  = DIVIDE ([Total Sales], [Active Customers], 0)
19 Orders per Customer   = DIVIDE ([Order Count], [Active Customers], 0)
20 Repeat Purchase Rate  =
21 VAR CustomersWithOrders =
22     SUMMARIZE (Fact_Sales, Fact_Sales[CustomerKey], "OrderCnt", [Order Count])
23 VAR RepeatCustomers =
24     COUNTROWS (FILTER (CustomersWithOrders, [OrderCnt] > 1))
25 RETURN
26     DIVIDE (RepeatCustomers, [Active Customers], 0)

```

Key design: New Customers is the most complex measure in this category. It builds a virtual table (ADDCOLUMNS) containing every customer and their first-ever order date, then FILTERs to those whose first order falls within the current date context. The VAR/RETURN pattern keeps intermediate logic readable and debuggable.

Repeat Purchase Rate uses SUMMARIZE to build a customer-order-count table, filters for customers

with more than one order, and divides by total active customers. This measure reveals customer retention quality across segments and time periods.

Geography & Market Analysis Measures

```

1 Market Sales % =
2     DIVIDE (
3         CALCULATE([Total Sales], ALLEXCEPT(DimGeography, DimGeography[Market])),
4         CALCULATE([Total Sales], ALL(DimGeography)),
5         0
6     )
7
8 Regional Profitability =
9     CALCULATE([Profit Margin], ALLEXCEPT(DimGeography, DimGeography[Region]))

```

Key design: Both measures use ALLEXCEPT to preserve filtering at their respective level (Market or Region) while removing lower-level geography filters. This ensures correct contribution calculations even when a City or State is selected.

Shipping Analysis Measures

```

1 Avg Shipping Days      = AVERAGE(Fact_Sales[ShippingDays])
2 Avg Shipping Cost      = AVERAGE(Fact_Sales[ShippingCost])
3 Shipping Cost per Unit = DIVIDE([Total Shipping Cost], [Total Quantity], 0)
4
5 On-Time Ship Rate      =
6 VAR TotalOrders = [Order Count]
7 VAR ExpressOrders =
8     CALCULATE([Order Count], DimShipping[ShippingSpeed] = "Express")
9 RETURN
10    DIVIDE(ExpressOrders, TotalOrders, 0)

```

Key design: On-Time Ship Rate defines "on-time" as express shipping methods. The CALCULATE with a direct filter on DimShipping[ShippingSpeed] activates the existing relationship between DimShipping and Fact_Sales, making this measure context-aware.

Dynamic KPIs — Performance vs Targets

```

1 KPI Sales vs Target =
2     VAR TargetAmount = 6000000
3     RETURN DIVIDE([Total Sales], TargetAmount, 0)
4
5 KPI Profit vs Target =
6     VAR TargetProfit = 500000
7     RETURN DIVIDE([Total Profit], TargetProfit, 0)
8
9 YTD Status =
10    IF([Sales YTD] > [Sales PY], "|_Above_Last_Year", "|_Below_Last_Year")

```

Report Design and Findings

The report uses a custom navy-and-coral theme with white card backgrounds and consistent typography. A single Year slicer on the Dashboard is synced across all four pages via Power BI's slicer sync feature — changing the year on any page updates every visual in the report.

Page 1: Executive Dashboard

Designed for a C-level audience seeking a 30-second business pulse check.

Visual	Configuration	Insight
KPI Cards (4)	Sales: \$5.63M, Profit: \$478.9K, Customers: 4,867, Margin: 8.5%	Instant health snapshot; all key metrics visible without interaction
Line Chart	Monthly Sales + Sales PY (dual series, year-level drill-down)	Clear Q4 seasonal peaks across all four years; 2014 shows the strongest year-end
Donut Chart	Sales by Category	Office Supplies = 49%, Technology = 29%, Furniture = 22% — Technology punches above its volume with higher margins
Year Slicer	Vertical list, multi-select, synced to all pages	Single filter entry point for the entire report

Table 2: Executive Dashboard visual layout

Page 2: Product Analysis

Designed for product managers and category owners seeking portfolio-level insights.

Visual	Configuration	Insight
Clustered Bar Chart	Sales + Profit by Category, dual axis	Technology generates the highest profit despite 37% lower volume than Office Supplies
Table (Drill-down)	Category → SubCategory; columns: Sales, Profit, Margin, Category Share %	Reveals sub-category winners — Phones and Chairs drive disproportionate profit within their categories
Scatter Chart	X: Avg Unit Price, Y: Profit Margin, Legend: Category	Technology products cluster in the premium quadrant (high price, high margin); Office Supplies cluster in the commodity quadrant

Table 3: Product Analysis page visual layout

Page 3: Customer & Geography

Designed for regional sales managers and marketing strategists.

Visual	Configuration	Insight
Bing Map	Bubble size = Total Sales, Location = Country	APAC, LATAM, and the US dominate. European sales are spread evenly across countries rather than concentrated
Market Table	Customers, Sales, Revenue per Customer, Orders per Customer	US leads in revenue per customer (\$1,157); APAC has the highest raw customer count
Bar Chart	Sales + Profit by Region	Central and South regions are the strongest performers; EMEA and Africa show growth potential

Table 4: Customer & Geography page visual layout

Page 4: Shipping & Time Trends

Designed for operations and logistics stakeholders.

Visual	Configuration	Insight
Dual Line Chart	Sales by Order Date vs Sales by Ship Date	A 3–5 day lag between order and ship dates is visible — revenue recognition timing shifts by roughly one calendar week
Gauge	Avg Shipping Days (4.0), Target needle at 3.0 days	Standard Class (4.2 days average) is the main contributor to the gap; Same Day shipping meets the target
Bar Chart	Order Count by Ship Mode	Standard Class accounts for 59.5% of all orders — the dominant customer preference

Table 5: Shipping & Time Trends page visual layout

Conclusion

The Sample Superstore dataset was transformed from a flat 25,035-row CSV into a fully normalised star-schema model with 38 DAX measures across seven analytical categories, delivered as a four-page interactive Power BI report. The project followed a structured pipeline: systematic data cleaning with Python (pandas), dimensional modelling with surrogate keys and an inactive relationship for ship-date analysis, and DAX measure authoring covering base aggregations, profitability ratios, time intelligence, product, customer, geographic, and shipping analytics.

Key analytical outcomes include:

- Overall **profit margin of 8.5%** with significant category variation — Technology outperforms Furniture by approximately $2.1\times$.
- **Shipping cost represents 10.8% of sales**, a material operational cost warranting continued monitoring.
- **Consumer segment dominates** with 51% of orders, followed by Corporate (30%) and Home Office (18%).
- Time-intelligence measures revealed consistent year-over-year growth across most categories, with the Technology segment showing the strongest upward trend.

The final Power BI report provides executive-level visibility into sales performance, profitability, customer behaviour, and operational efficiency through a cohesive navy-and-coral themed interface with synced slicers, drill-down hierarchies, and dynamic KPI targets.